

KM44C1000B**CMOS DRAM****1M × 4 Bit CMOS Dynamic RAM with Fast Page Mode****FEATURES**• **Performance range:**

	t_{RAC}	t_{CAC}	t_{RC}
KM44C1000B-6	60ns	15ns	110ns
KM44C1000B-7	70ns	20ns	130ns
KM44C1000B-8	80ns	20ns	150ns

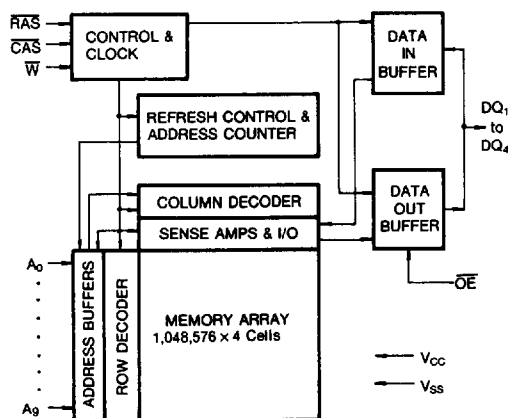
- **Fast Page Mode operation**
- **CAS-before-RAS refresh capability**
- **RAS-only and Hidden Refresh capability**
- **Fast parallel test mode capability**
- **TTL compatible inputs and outputs**
- **Early write or output enable controlled write**
- **Single +5V ± 10% power supply**
- **1024 cycles/16ms refresh**
- **JEDEC standard pinout**
- **Available in plastic DIP, SOJ, ZIP and TSOP packages**

GENERAL DESCRIPTION

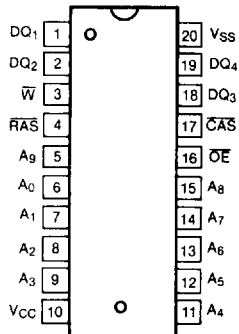
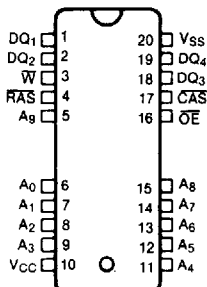
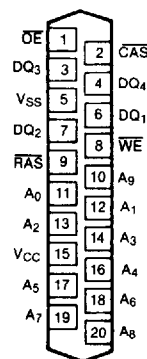
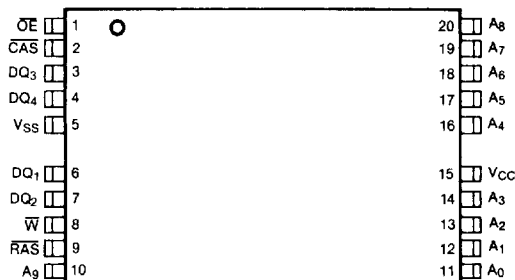
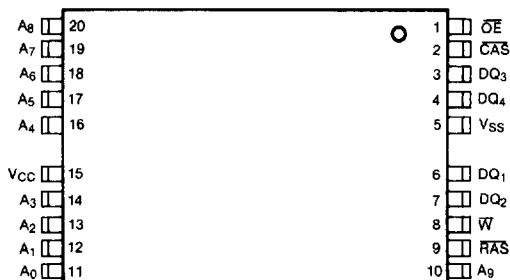
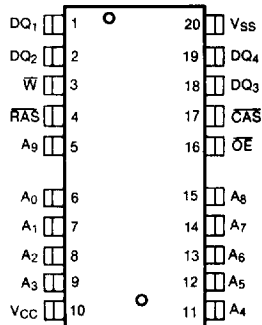
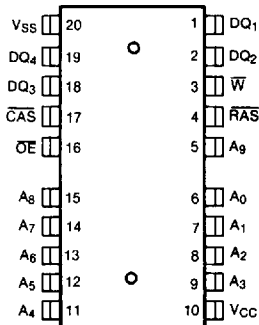
The Samsung KM44C1000B is a high speed CMOS 1,048,516 × 4 Dynamic Random Access Memory. Its design is optimized for high performance applications such as mainframes and mini computers, graphics and high performance microprocessor systems.

The KM44C1000B features Fast Page Mode operation which allows high speed random access of memory cells within the same row. CAS-before-RAS refresh capability provides on-chip auto refresh as an alternative to RAS-only Refresh. All inputs and output are fully TTL compatible.

The KM44C1000B is fabricated using Samsung's advanced CMOS process.

FUNCTIONAL BLOCK DIAGRAM**ORDERING INFORMATION**

Part No.	Access Time	Package
KM44C1000BP-6	60 ns	300 mil, 20 DIP
KM44C1000BP-7	70 ns	
KM44C1000BP-8	80 ns	
KM44C1000BJ-6	60 ns	300 mil, 20 SOJ
KM44C1000BJ-7	70 ns	
KM44C1000BJ-8	80 ns	
KM44C1000BZ-6	60 ns	400 mil, 20 ZIP
KM44C1000BZ-7	70 ns	
KM44C1000BZ-8	80 ns	
KM44C1000BV-6	60 ns	20 TSOP (I) (Forward)
KM44C1000BV-7	70 ns	
KM44C1000BV-8	80 ns	
KM44C1000BVR-6	60 ns	20 TSOP (I) (Reverse)
KM44C1000BVR-7	70 ns	
KM44C1000BVR-8	80 ns	
KM44C1000BT-6	60 ns	20 TSOP (II) (Forward)
KM44C1000BT-7	70 ns	
KM44C1000BT-8	80 ns	
KM44C1000BTR-6	60 ns	20 TSOP (II) (Reverse)
KM44C1000BTR-7	70 ns	
KM44C1000BTR-8	80 ns	

KM44C1000B**CMOS DRAM****PIN CONFIGURATION (Top Views)****• KM44C1000BP****• KM44C1000BJ****• KM44C1000BZ****• KM44C1000BV****• KM44C1000BVR****• KM44C1000BT****• KM44C1000BTR**

Pin Names	Pin Function
A ₀ -A ₉	Address Inputs
RAS	Row Address Strobe
CAS	Column Address Strobe
W	Read/Write Input
OE	Data Output Enable
DQ ₁ /DQ ₄	Data In/Data Out
V _{CC}	Power (+ 5V)
V _{SS}	Ground

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ABSOLUTE MAXIMUM RATINGS*

Parameter	Symbol	Value	Units
Voltage on Any Pin Relative to V_{SS}	V_{IN}, V_{OUT}	-1 to +7.0	V
Voltage on V_{CC} Supply Relative to V_{SS}	V_{CC}	-1 to +7.0	V
Storage Temperature	T_{stg}	-55 to +150	°C
Power Dissipation	P_D	600	mW
Short Circuit Output Current	I_{OS}	50	mA

* Permanent device damage may occur if "ABSOLUTE MAXIMUM RATINGS" are exceeded. Functional Operation should be restricted to the conditions as detailed in the operational sections of this data sheet. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

RECOMMENDED OPERATING CONDITIONS (Voltage reference to V_{SS} , $T_A=0$ to 70°C)

Parameter	Symbol	Min	Typ	Max	Unit
Supply Voltage	V_{CC}	4.5	5.0	5.5	V
Ground	V_{SS}	0	0	0	V
Input High Voltage	V_{IH}	2.4	—	$V_{CC}+1$	V
Input Low Voltage	V_{IL}	-1.0	—	0.8	V

DC AND OPERATING CHARACTERISTICS

(Recommended operating conditions unless otherwise noted)

Parameter		Symbol	Min	Max	Unit
Operating Current* ($\overline{RAS}$, $\overline{CAS}$, Address Cycling @ $t_{RC}=\min$)	KM44C1000B-6	I_{CC1}	—	90	mA
	KM44C1000B-7		—	80	mA
	KM44C1000B-8		—	70	mA
Standby Current ($\overline{RAS}=\overline{CAS}=V_{IH}$)		I_{CC2}	—	2	mA
$\overline{RAS}$ -Only Refresh Current* ($\overline{CAS}=V_{IH}$, $\overline{RAS}$, Address Cycling @ $t_{RC}=\min$.)	KM44C1000B-6	I_{CC3}	—	90	mA
	KM44C1000B-7		—	80	mA
	KM44C1000B-8		—	70	mA
Fast Page Mode Current* ($\overline{RAS}=V_{IL}$, $\overline{CAS}$, Address Cycling @ $t_{PC}=\min$.)	KM44C1000B-6	I_{CC4}	—	70	mA
	KM44C1000B-7		—	60	mA
	KM44C1000B-8		—	50	mA
Standby Current ($\overline{RAS}=\overline{CAS}=V_{CC}-0.2V$)		I_{CC5}	—	1	mA
$\overline{CAS}$ -Before- $\overline{RAS}$ Refresh Current* ($\overline{RAS}$, $\overline{CAS}$ Cycling @ $t_{RC}=\min$.)	KM44C1000B-6	I_{CC6}	—	90	mA
	KM44C1000B-7		—	80	mA
	KM44C1000B-8		—	70	mA
Standby Current ($\overline{RAS}=V_{IH}$, $\overline{CAS}=V_{IL}$, $\overline{DOUT}$ Enable)		I_{CC7}	—	5	mA
Input Leakage Current (Any input $0 \leq V_{IN} \leq 6.5V$, all other pins not under test = 0 volts)		I_{IL}	-10	10	μA
Output Leakage Current (Data out is disabled, $0 \leq V_{OUT} \leq 5.5V$)		I_{OL}	-10	10	μA
Output High Voltage Level ($I_{OH}=-5mA$)		V_{OH}	2.4	—	V
Output Low Voltage Level ($I_{OL}=4.2mA$)		V_{OL}	—	0.4	V

*NOTE: I_{CC1} , I_{CC3} , I_{CC4} , I_{CC6} are dependent on output loading and cycle rates. Specified value are obtained with the output open. I_{CC} is specified as an average current. In I_{CC1} , I_{CC3} address transition should be changed maximum two while $\overline{RAS}=V_{IL}$. In I_{CC4} , address transition should be changed only once while $\overline{CAS}=V_{IH}$.

KM44C1000B**CMOS DRAM****CAPACITANCE** ($T_A = 25^\circ\text{C}$)

Parameter	Symbol	Min	Max	Unit
Input Capacitance (A_0 - A_9)	C_{IN1}	—	5	pF
Input Capacitance ($\overline{RAS}$, $\overline{CAS}$, $\overline{W}$, $\overline{OE}$)	C_{IN2}	—	7	pF
Output Capacitance (DQ_1 - DQ_4)	C_{DQ}	—	7	pF

AC CHARACTERISTICS ($0^\circ\text{C} \leq T_A \leq 70^\circ\text{C}$, $V_{CC} = 5.0\text{V} \pm 10\%$, See notes 1,2)

Parameter	Symbol	KM44C1000B-6		KM44C1000B-7		KM44C1000B-8		Units	Notes
		Min	Max	Min	Max	Min	Max		
Random read or write cycle time	t_{RC}	110		130		150		ns	
Read-modify-write cycle time	t_{RWC}	155		185		205		ns	
Access time from $\overline{RAS}$	t_{RAC}		60		70		80	ns	3,4,11
Access time from $\overline{CAS}$	t_{CAC}		15		20		20	ns	3,4,5
Access time from column address	t_{AA}		30		35		40	ns	3,11
$\overline{CAS}$ to output in Low-Z	t_{CLZ}	0		0		0		ns	3
Output buffer turn-off delay	t_{OFF}	0	15	0	20	0	20	ns	7
Transition time (rise and fall)	t_r	3	50	3	50	3	50	ns	2
$\overline{RAS}$ precharge time	t_{RP}	40		50		60		ns	
$\overline{RAS}$ pulse width	t_{RAS}	60	10,000	70	10,000	80	10,000	ns	
$\overline{RAS}$ hold time	t_{RSH}	15		20		20		ns	
$\overline{CAS}$ hold time	t_{CSH}	60		70		80		ns	
$\overline{CAS}$ pulse width	t_{CAS}	15	10,000	20	10,000	20	10,000	ns	
$\overline{RAS}$ to $\overline{CAS}$ delay time	t_{RCD}	20	45	20	50	20	60	ns	4
$\overline{RAS}$ to column address delay time	t_{RAD}	15	30	15	35	15	40	ns	11
$\overline{CAS}$ to $\overline{RAS}$ precharge time	t_{CRP}	5		5		5		ns	
Row address set-up time	t_{ASR}	0		0		0		ns	
Row address hold time	t_{RAH}	10		10		10		ns	
Column address set-up time	t_{ASC}	0		0		0		ns	
Column address hold time	t_{CAH}	15		15		15		ns	
Column address hold time referenced to $\overline{RAS}$	t_{AR}	50		55		60		ns	6
Column Address to $\overline{RAS}$ lead time	t_{RAL}	30		35		40		ns	
Read command set-up time	t_{RCS}	0		0		0		ns	
Read command hold time referenced to $\overline{CAS}$	t_{RCH}	0		0		0		ns	9
Read command hold time referenced to $\overline{RAS}$	t_{RRH}	0		0		0		ns	9
Write command hold time	t_{WCH}	10		15		15		ns	
Write command hold time referenced to $\overline{RAS}$	t_{WCR}	45		55		60		ns	6
Write command pulse width	t_{WP}	10		15		15		ns	
Write command to $\overline{RAS}$ lead time	t_{RWL}	15		20		20		ns	
Write command to $\overline{CAS}$ lead time	t_{CWL}	15		20		20		ns	
Data-in set-up time	t_{DS}	0		0		0		ns	10

KM44C1000B**CMOS DRAM****AC CHARACTERISTICS** (Continued)

Parameter	Symbol	KM44C1000B-6		KM44C1000B-7		KM44C1000B-8		Units	Notes
		Min	Max	Min	Max	Min	Max		
Data-in hold time	t _{DH}	15		15		15		ns	10
Data-in hold time referenced to $\overline{\text{RAS}}$	t _{DHR}	50		55		60		ns	6
Refresh period (1,024 cycles)	t _{REF}		16		16		16	ms	
Write command set-up time	t _{WCS}	0		0		0		ns	8
$\overline{\text{CAS}}$ to $\overline{\text{W}}$ delay time	t _{CWD}	40		50		50		ns	8
$\overline{\text{RAS}}$ to $\overline{\text{W}}$ delay time	t _{RWD}	85		100		110		ns	8
Column address to $\overline{\text{W}}$ delay time	t _{AWD}	55		65		70		ns	8
$\overline{\text{CAS}}$ set-up time ($\overline{\text{CAS}}$ -before- $\overline{\text{RAS}}$ refresh)	t _{CSR}	10		10		10		ns	
$\overline{\text{CAS}}$ hold time ($\overline{\text{CAS}}$ -before- $\overline{\text{RAS}}$ refresh)	t _{CHR}	10		15		15		ns	
$\overline{\text{RAS}}$ to $\overline{\text{CAS}}$ precharge time	t _{RPC}	5		5		5		ns	
$\overline{\text{CAS}}$ precharge time ($\overline{\text{C-B-R}}$ counter test cycle)	t _{CPT}	20		25		30		ns	
Access time from $\overline{\text{CAS}}$ precharge	t _{CPA}		35		40		45	ns	3
Fast page mode cycle time	t _{PC}	40		45		50		ns	
Fast page mode read-modify-write cycle time	t _{PRWC}	80		95		100		ns	
$\overline{\text{RAS}}$ pulse width (Fast Page Mode)	t _{RASP}	60	200,000	70	200,000	80	200,000	ns	
$\overline{\text{CAS}}$ precharge time (Fast page mode)	t _{CP}	10		10		10		ns	
$\overline{\text{RAS}}$ hold time from $\overline{\text{CAS}}$ precharge	t _{RHCP}	35		40		45		ns	
$\overline{\text{OE}}$ access time	t _{OEA}		15		20		20	ns	
$\overline{\text{OE}}$ to data delay	t _{OED}	15		20		20		ns	
Output buffer turn off delay time from $\overline{\text{OE}}$	t _{OEZ}	0	15	0	20	0	20	ns	
$\overline{\text{OE}}$ command hold time	t _{OEH}	15		20		20		ns	
Write command set-up time (test mode in)	t _{WTS}	10		10		10		ns	
Write command hold time (test mode in)	t _{WTH}	10		10		10		ns	
$\overline{\text{W}}$ to $\overline{\text{RAS}}$ precharge time ($\overline{\text{C-B-R}}$ cycle)	t _{WRP}	10		10		10		ns	
$\overline{\text{W}}$ to $\overline{\text{RAS}}$ hold time ($\overline{\text{C-B-R}}$ cycle)	t _{WRH}	10		10		10		ns	

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TEST MODE CYCLE

(Note. 12)

Parameter	Symbol	KM44C1000B-6		KM44C1000B-7		KM44C1000B-8		Units	Notes
		Min	Max	Min	Max	Min	Max		
Random read or write cycle time	t_{RC}	115		135		155		ns	
Read-modify-write cycle time	t_{RWC}	160		190		210		ns	
Access time from $\overline{RAS}$	t_{RAC}		65		75		85	ns	3,4,11
Access time from $\overline{CAS}$	t_{CAC}		20		25		25	ns	3,4,5
Access time from column address	t_{AA}		35		40		45	ns	3,11
$\overline{RAS}$ pulse width	t_{RAS}	65	10,000	75	10,000	85	10,000	ns	
$\overline{CAS}$ pulse width	t_{CAS}	20	10,000	25	10,000	25	10,000	ns	
$\overline{RAS}$ hold time	t_{RSH}	20		25		25		ns	
$\overline{CAS}$ hold time	t_{CSH}	65		75		85		ns	
Column address to $\overline{RAS}$ lead time	t_{RAL}	35		40		45		ns	
$\overline{CAS}$ to write enable delay	t_{CWD}	45		55		55		ns	8
$\overline{RAS}$ to write enable delay	t_{RWD}	90		105		115		ns	8
Column address to $\overline{W}$ delay time	t_{AWD}	60		70		75		ns	8
Fast mode cycle time	t_{PC}	45		50		55		ns	
Fast page mode read-modify-write	t_{PRWC}	85		100		105		ns	
$\overline{RAS}$ pulse width (Fast page mode)	t_{RASP}	65	200,000	75	200,000	85	200,000	ns	
Access time from $\overline{CAS}$ precharge	t_{CPA}		40		45		50	ns	3
$\overline{OE}$ access time	t_{OEA}		20		25		25	ns	
$\overline{OE}$ to data delay	t_{OED}	20		25		25		ns	
$\overline{OE}$ command hold time	t_{OEH}	20		25		25		ns	

NOTES

1. An initial pause of 200 μ s is required after power up followed by any 8 $\overline{CAS}$ -before- $\overline{RAS}$ or $\overline{RAS}$ -only Refresh cycles before proper device operation is achieved.
2. $V_{IH(min)}$ and $V_{IL(max)}$ are reference levels for measuring timing of input signals. Transition times are measured between $V_{IH(min)}$ and $V_{IL(max)}$, and are assumed to be 5ns for all inputs.
3. Measured with a load equivalent to 2 TTL loads and 100pF.
4. Operation within the $t_{RCD(max)}$ limit insures that $t_{RAC(max)}$ can be met. $t_{RCD(max)}$ is specified as a reference point only. If t_{RCD} is greater than the specified $t_{RCD(max)}$ limit, then access time is controlled exclusively by t_{CAC} .
5. Assumes that $t_{RCD} \geq t_{RCD(max)}$.
6. t_{AR} , t_{WCR} , t_{DHR} are referenced to $t_{RAD(max)}$.
7. This parameter defines the time at which the output achieves the open circuit condition and is not referenced to V_{OH} or V_{OL} .
8. t_{WCS} , t_{RWD} , t_{CWD} and t_{AWD} are non restrictive operating parameters. They are included in the data sheet as electrical characteristics only. If $t_{WCS} \geq t_{WCS(min)}$ the cycle is an early write cycle and the data out pin will remain high impedance for the duration of the cycle. If $t_{CWD} \geq t_{CWD(min)}$ and $t_{RWD} \geq t_{RWD(min)}$ and $t_{AWD} \geq t_{AWD(min)}$, then the cycle is a read-modify-write cycle and the data out will contain the data read from the selected address. If neither of the above conditions are satisfied, the condition of the data out is indeterminate.
9. Either t_{RCH} or t_{RRH} must be satisfied for a read cycle.
10. These parameters are referenced to the $\overline{CAS}$ leading edge in early write cycles and to the $\overline{W}$ leading edge in read-modify-write cycles.
11. Operation within the $t_{RAD(max)}$ limit insures that $t_{RAC(max)}$ can be met. $t_{RAD(max)}$ is specified as a reference point only. If t_{RAD} is greater than the specified $t_{RAD(max)}$ limit, then access time is controlled by t_{AA} .
12. These specifications are applied in the test mode.
13. In test mode read cycle, the value of t_{RAC} , t_{AA} , t_{CAC} is delayed by 2ns to 5ns for the specified value. These parameters should be specified in test mode cycles by adding the above value to the specified value in this data sheet.
14. $t_{OFF(max)}$ and $t_{OEZ(max)}$ define the time at which the output achieves the open circuit condition and are not referenced to output voltage level.

CMOS DRAM

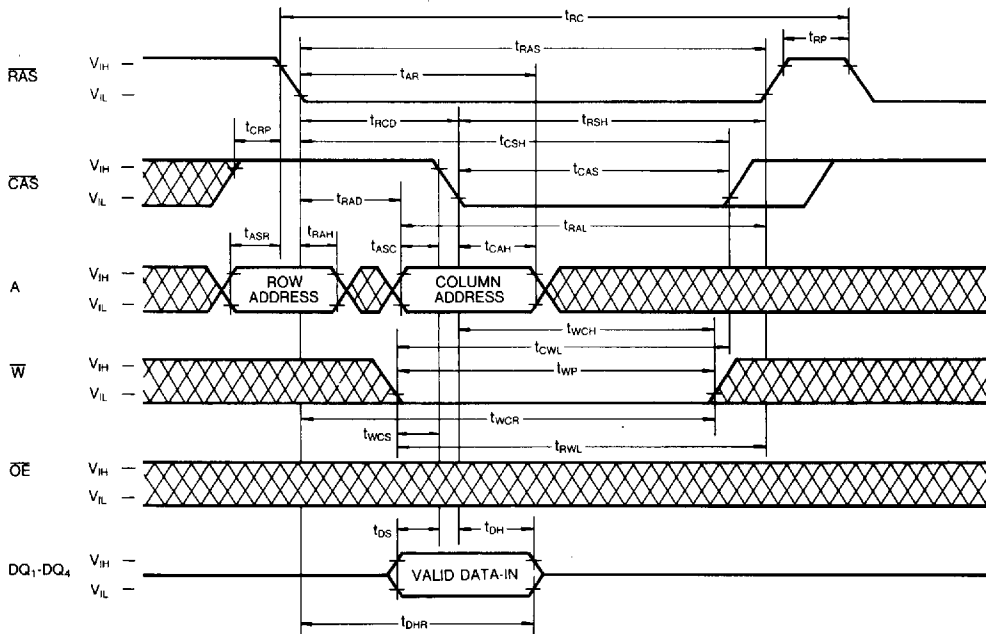
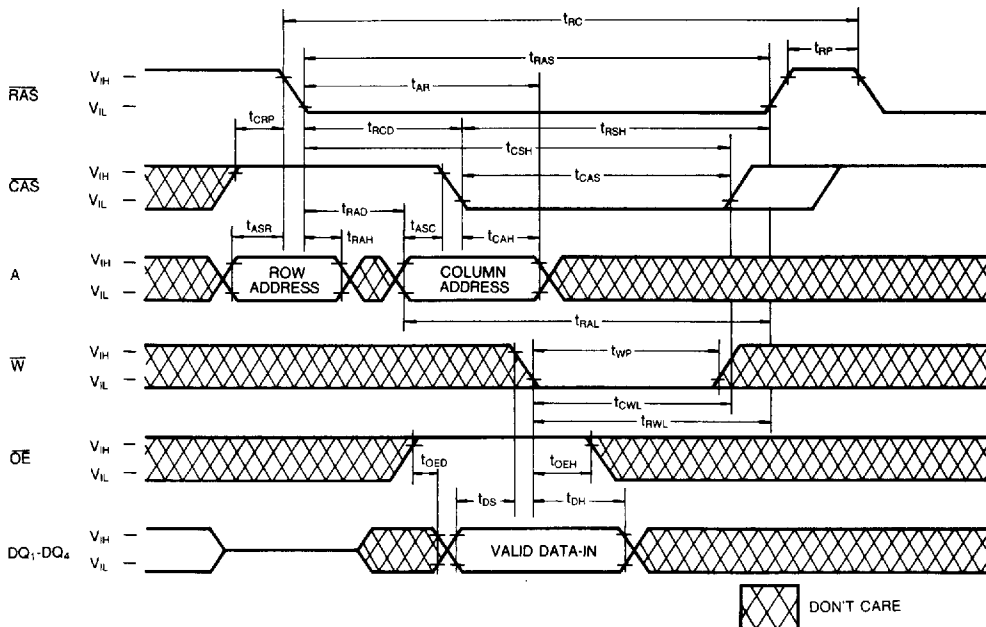
 DON'T CARE

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TIMING DIAGRAMS (Continued)

WRITE CYCLE (EARLY CYCLE)

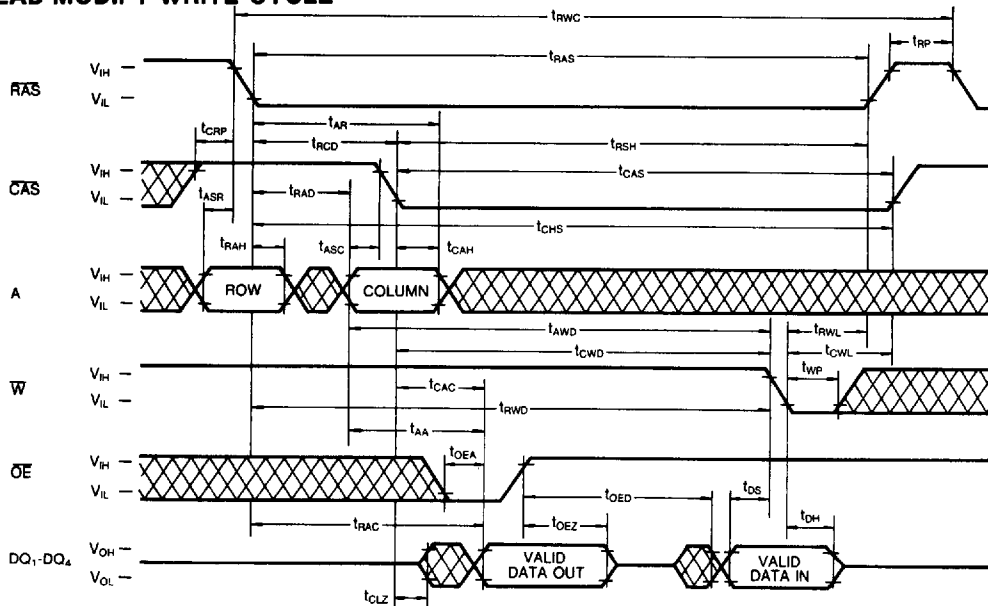
WRITE CYCLE ($\overline{OE}$ CONTROLLED WRITE)

KM44C1000B

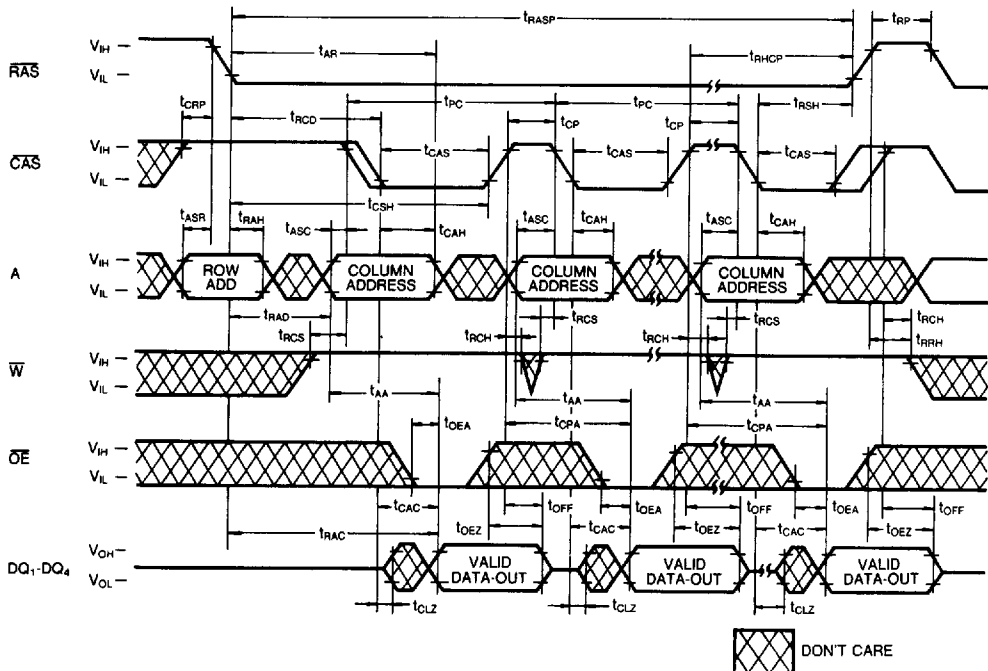
CMOS DRAM

TIMING DIAGRAMS (Continued)

READ-MODIFY-WRITE CYCLE



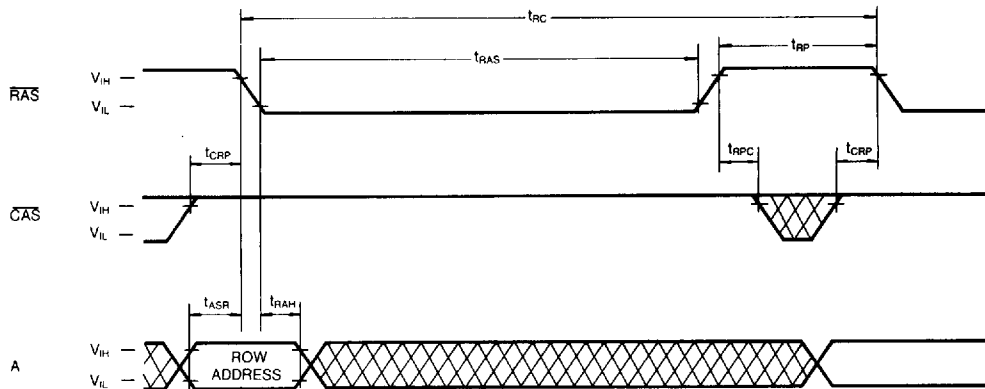
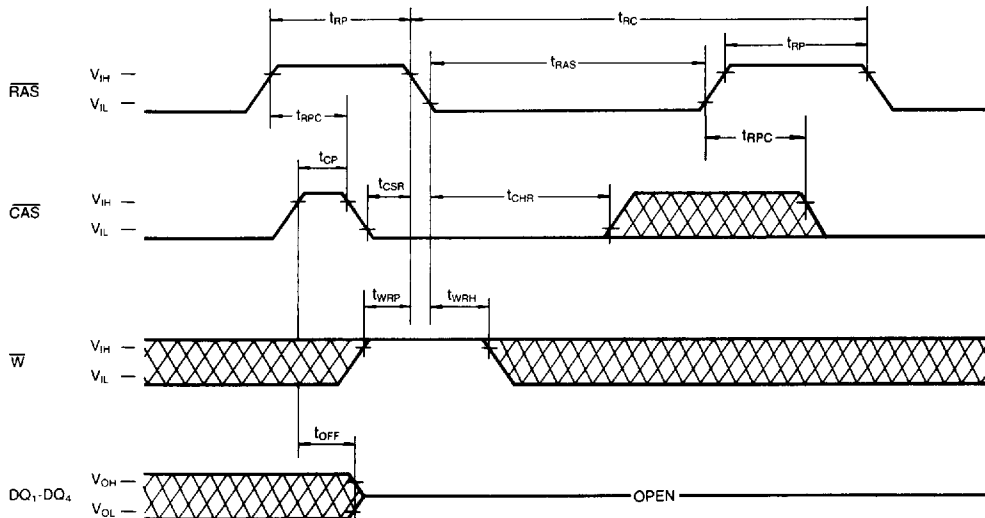
FAST PAGE MODE READ CYCLE



CMOS DRAM

FAST PAGE MODE WRITE CYCLE (EARLY WRITE)



KM44C1000B**CMOS DRAM****TIMING DIAGRAMS** (Continued) **$\overline{\text{RAS}}$ ONLY REFRESH CYCLE**Note: $\overline{\text{W}}$, $\overline{\text{OE}}$ =Don't Care **$\overline{\text{CAS}}$ -BEFORE- $\overline{\text{RAS}}$ REFRESH CYCLE**Note: $\overline{\text{OE}}$, Address=Don't Care

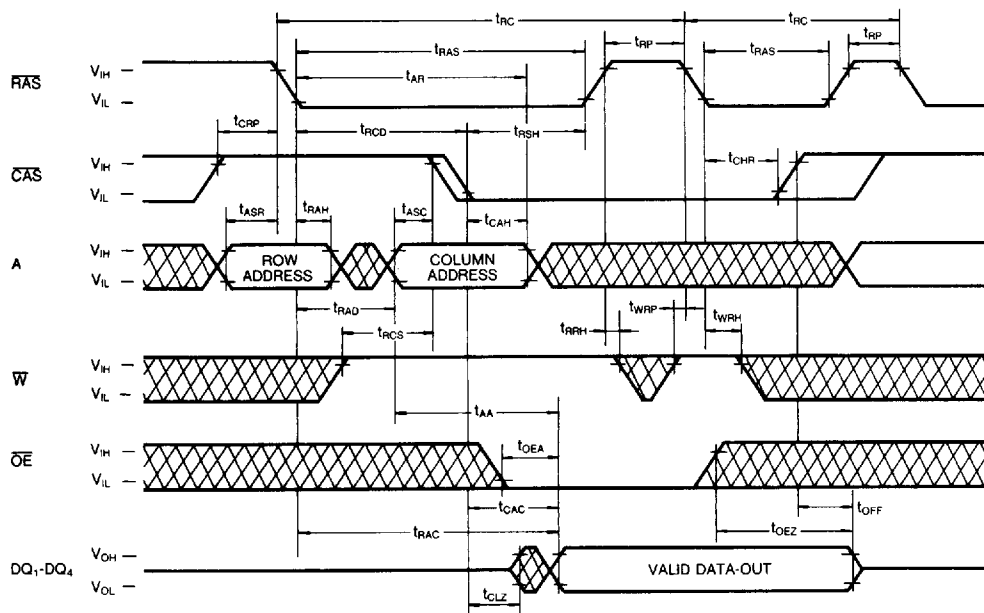
DON'T CARE

KM44C1000B

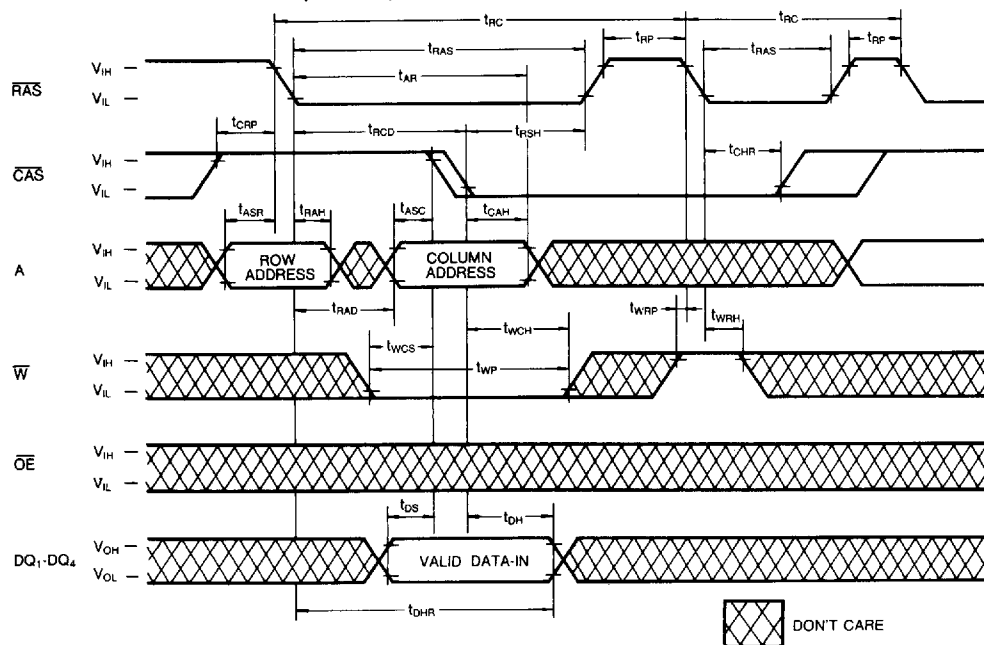
CMOS DRAM

TIMING DIAGRAMS (Continued)

HIDDEN REFRESH CYCLE (READ)

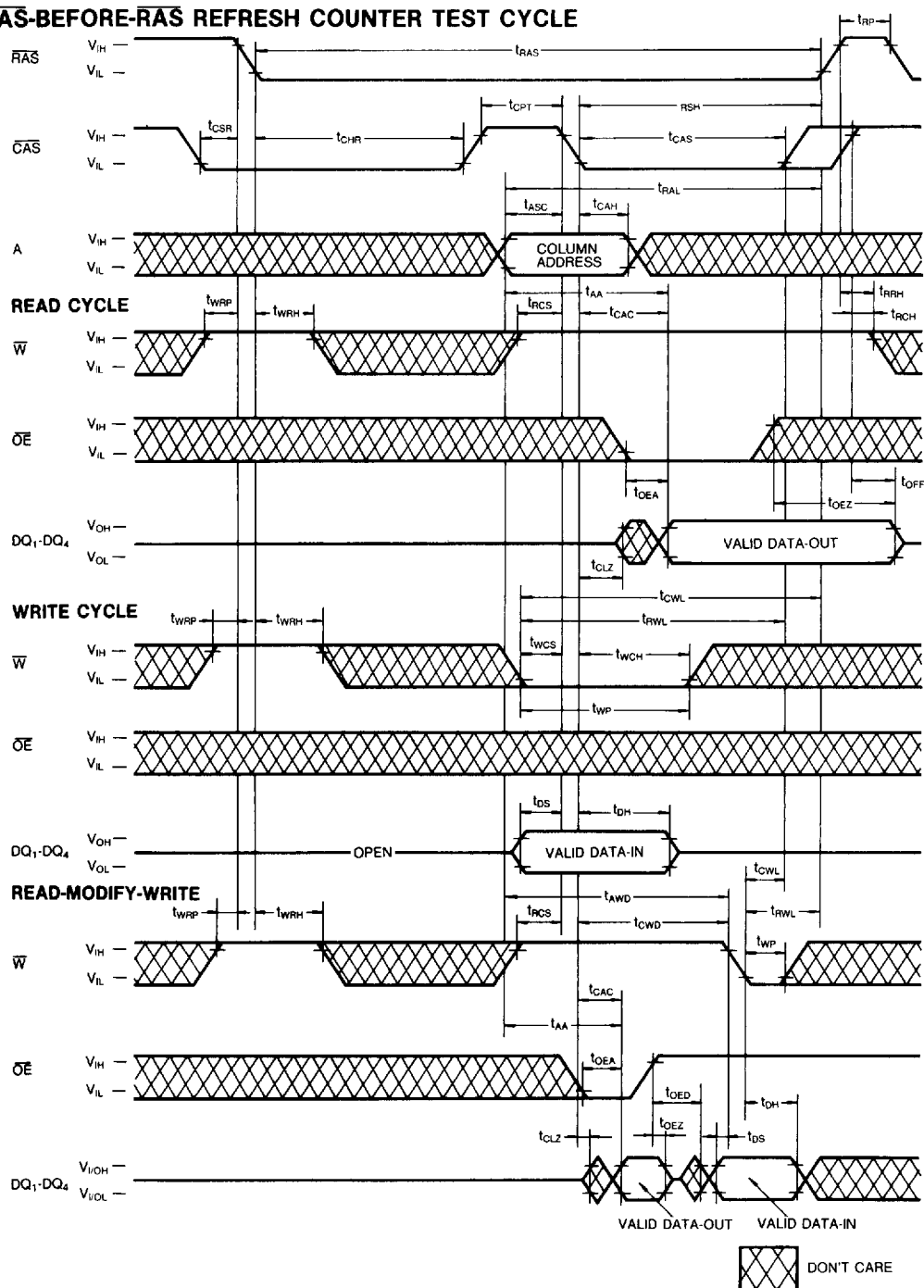


HIDDEN REFRESH CYCLE (WRITE)



TIMING DIAGRAMS (Continued)

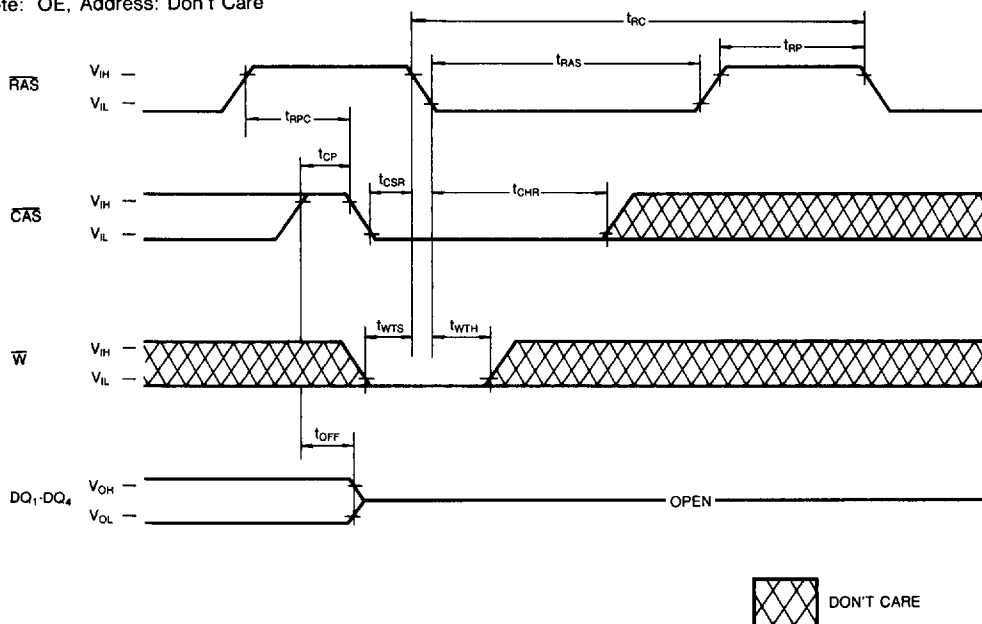
CAS-BEFORE-RAS REFRESH COUNTER TEST CYCLE



TIMING DIAGRAMS (Continued)

TEST MODE IN CYCLE

Note: $\overline{OE}$, Address: Don't Care



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TEST MODE DESCRIPTION

The KM44C1000B is the RAM organized 1,048,576 words by 4 bit, it is internally organized 524,288 words by 8 bits. In "Test Mode," data are written into 8 sectors in parallel and retrieved the same way. Column address bit A_0 is not used. If, upon reading, two bits on one I/O pin are equal (all "1" or "0"s) the I/O pin indicates a "1." If they were not equal, the I/O pin would indicate a "0." In "Test

Mode," the 1M $\times$ 4 DRAM can be tested as if it were a 512K $\times$ 4 DRAM. $\overline{W}$, $\overline{CAS}$ -BEFORE- $\overline{RAS}$ Cycle (Test Mode in Cycle) puts the device into "Test Mode." And " $\overline{CAS}$ -BEFORE- $\overline{RAS}$ REFRESH CYCLE" or " $\overline{RAS}$ -only Refresh Cycle" puts it back into "Normal Mode." The "Test Mode" function reduces test time (1/2 in cases of N test pattern).

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DEVICE OPERATION

The KM44C1000B contains 4,194,304 memory locations. Twenty address bits are required to address a particular 4-bit word in the memory array. Since the KM44C1000B has only 10 address input pins, time multiplexed addressing is used to input 10 row (A_0 - A_9) and 10 column (A_{10} - A_{19}) addresses. The multiplexing is controlled by the timing relationship between the row address strobe ($\overline{RAS}$), the column address strobe ($\overline{CAS}$) and the valid row and column address inputs.

Operating of the KM44C1000B begins by strobing in a valid row address with $\overline{RAS}$ while $\overline{CAS}$ remains high. Then the address on the 10 address input pins (A_0 - A_9) is changed from a row address to a column address and is strobed in by $\overline{CAS}$. This is the beginning of any KM44C1000B cycle in which a memory location is accessed. The specific type of cycle is determined by the state of the write enable pin and various timing relationships. The cycle is terminated when both $\overline{RAS}$ and $\overline{CAS}$ have returned to the high state. Another cycle can be initiated after $\overline{RAS}$ remains high long enough to satisfy the $\overline{RAS}$ precharge time (t_{RP}) requirement.

 $\overline{RAS}$ and $\overline{CAS}$ Timing

The minimum $\overline{RAS}$ and $\overline{CAS}$ pulse widths are specified by $t_{RAS(min)}$ and $t_{CAS(min)}$ respectively. These minimum pulse widths must be satisfied for proper device operation and data integrity. Once a cycle is initiated by bringing $\overline{RAS}$ low, it must not be aborted prior to satisfying the minimum $\overline{RAS}$ and $\overline{CAS}$ pulse widths. In addition, a new cycle must not begin until the minimum $\overline{RAS}$ precharge time, t_{RP} , has been satisfied. Once a cycle begins, internal clocks and other circuits within the KM44C1000B begin a complex sequence of events. If the sequence is broken by violating minimum timing requirements, loss of data integrity can occur.

Read

A read cycle is achieved by maintaining the write enable input ($\overline{WE}$) high during a $\overline{RAS}/\overline{CAS}$ cycle. The access time is normally specified with respect to the falling edge of $\overline{RAS}$. But the access time also depends on the falling edge of $\overline{CAS}$ and on the valid column address transition.

If $\overline{CAS}$ goes low before $t_{RCD(max)}$ and if the column address is valid before $t_{RAC(max)}$, then the access time to valid data is specified by $t_{RAC(min)}$. However, if $\overline{CAS}$ goes low after $t_{RCD(max)}$ or if the column address becomes valid after $t_{RAC(max)}$, access is specified by t_{CAC} or t_{AA} . In order to achieve the minimum access time, $t_{RAC(min)}$, it is necessary to meet both $t_{RCD(max)}$ and $t_{RAD(max)}$. The KM44C1000B has common data I/O pins.

This is the reason an output enable control input ($\overline{OE}$) has been provided so the output buffer can be precisely controlled. For data to appear at the outputs, $\overline{OE}$ must be low for the period of time defined by t_{OEA} and t_{OEZ} .

Write

The KM44C1000B can perform early write, late write and read-modify-write cycles. The difference between these cycles is in the state of data-out and is determined by the timing relationship between $\overline{W}$, $\overline{OE}$ and $\overline{CAS}$. In any type of write cycle, Data-In must be valid at or before the falling edge of $\overline{W}$ or $\overline{CAS}$, whichever is later.

Early Write: An early write cycle is performed by bringing $\overline{W}$ low before $\overline{CAS}$. The data at the data input pins is written into the addressed memory cells. Throughout the early write cycle the output remains in the Hi-Z state. In the early write cycle the output buffers remain in the three state regardless of the state of the $\overline{OE}$ input.

Read-Modify-Write: In this cycle, valid data from the addressed cell appears at the output before and during the time that data is being written into the same cell location. This cycle is achieved by bringing $\overline{W}$ low after $\overline{CAS}$ and meeting the data sheet read-modify-write cycle timing requirements. This cycle requires using a separate I/O to avoid bus contention.

Late Write: If $\overline{W}$ is brought low after $\overline{CAS}$, a late write cycle will occur. The late write cycle is very similar to the read-modify-write cycle except that the timing parameters, t_{RWD} , t_{CWD} and t_{AWD} , are not necessarily met. The state of data-out is indeterminate since the output can be either Hi-Z or contain data depending on the timing conditions. This cycle requires a separate I/O to avoid bus contention.

Data Output

The KM44C1000B has a three-state output buffer which is controlled by $\overline{CAS}$ and $\overline{OE}$. Whenever $\overline{CAS}$ and $\overline{OE}$ are high (V_{IH}), the outputs are in the high impedance state. In any cycle in which valid data appears at the output the output goes into the low impedance state in a time specified by t_{CLZ} after the falling edge of $\overline{CAS}$. Invalid data may be present at the output during the time after t_{CLZ} and before the valid data appears at the output. The timing parameters t_{CAC} , t_{RAC} and t_{AA} specify when the valid data will be present at the output. The valid data remains at the output until $\overline{CAS}$ returns high. This is true even if a new $\overline{RAS}$ cycle occurs (as in hidden refresh). Each of the KM44C1000B operating cycles is listed below after the corresponding output state produced by the cycle.

KM44C1000B**CMOS DRAM****DEVICE OPERATION** (Continued)

Valid Output Data: Read, Read-Modify-Write, Hidden Refresh, Fast Page Mode Read, Fast Page Mode Read-Modify-Write.

Hi-Z Output State: Early Write, $\overline{RAS}$ -only Refresh, Fast Page Mode Write, $\overline{CAS}$ -before- $\overline{RAS}$ Refresh, $\overline{OE}$ controlled write.

Indeterminate Output State: Delayed Write (t_{CWD} or t_{RWD} are not met)

Refresh

The data in the KM44C1000B is stored on a tiny capacitor within each memory cell. Due to leakage the data may leak off after a period of time. To maintain data integrity it is necessary to refresh each of the rows every 16 ms. There are several ways to accomplish this.

$\overline{RAS}$ -Only Refresh: This is the most common method for performing refresh. It is performed by strobing in a row address with $\overline{RAS}$ while $\overline{CAS}$ remains high. This cycle must be repeated for each row.

$\overline{CAS}$ -before- $\overline{RAS}$ Refresh: The KM44C1000B has $\overline{CAS}$ -before- $\overline{RAS}$ on-chip refresh capability that eliminates the need for external refresh addresses. If $\overline{CAS}$ is held low for the specified set up time (t_{CSR}) before $\overline{RAS}$ goes low, the on-chip refresh circuitry is enabled. An internal refresh operation automatically occurs. The refresh address is supplied by the on-chip refresh address counter which is then internally incremented in preparation for the next $\overline{CAS}$ -before- $\overline{RAS}$ refresh cycle.

Hidden Refresh: A hidden refresh cycle may be performed while maintaining the latest valid data at the output by extending the $\overline{CAS}$ active time and cycling $\overline{RAS}$. The KM44C1000B hidden refresh cycle is actually a $\overline{CAS}$ -before- $\overline{RAS}$ refresh cycle within an extended read cycle. The refresh row address is provided by the on-chip refresh address counter.

Other Refresh Methods: It is also possible to refresh the KM44C1000B by using read, write or read-modify-write cycles. Whenever a row is accessed, all the cells in that row are automatically refreshed. There are certain applications in which it might be advantageous to perform refresh in this manner but in general $\overline{RAS}$ -only or $\overline{CAS}$ -before- $\overline{RAS}$ refresh is the preferred method.

Fast Page Mode

Fast page mode provides high speed read, write or read-modify-write access to all memory cells within a selected row. These cycles may be mixed in any order. A fast page mode cycle begins with a normal cycle. Then,

while $\overline{RAS}$ is kept low to maintain the $\overline{CAS}$ is cycled to strobe in additional column addresses. This eliminates the time required to set up and strobe sequential row addresses for the same page.

 $\overline{CAS}$ -before- $\overline{RAS}$ Refresh Counter**Test Cycle**

A special timing sequence using the $\overline{CAS}$ -before- $\overline{RAS}$ counter test cycle provides a convenient method of verifying the functionality of the $\overline{CAS}$ -before- $\overline{RAS}$ refresh activated circuitry.

After the $\overline{CAS}$ -before- $\overline{RAS}$ refresh operation, is $\overline{CAS}$ goes high and then low again while $\overline{RAS}$ is held low, the read and write operations are enabled.

This is shown in the $\overline{CAS}$ -before- $\overline{RAS}$ counter test cycle timing diagram. A memory cell can be addressed with 10 row address bits and 10 column address bits defined as follows:

Row Address—Bits A_0 through A_9 are supplied by the on-chip refresh counter.

Column Address — Bits A_0 through A_9 are supplied by the falling edge of $\overline{CAS}$ as in a normal memory cycle.

Suggested $\overline{CAS}$ -before- $\overline{RAS}$ Counter Test Procedure

The $\overline{CAS}$ -before- $\overline{RAS}$ refresh counter test cycle timing is used in each of the following steps:

1. Initialize the internal refresh counter by performing 8 $\overline{CAS}$ -before- $\overline{RAS}$ cycles.
2. Write a test pattern of "lows" into the memory cells at a single column address and 1024 row address. (The row addresses are supplied by the on-chip refresh counter).
3. Using read-modify-write cycles, read the "lows" written during step 2 and write "highs" into the same memory locations. Perform this step 1024 times so that highs are written into the 1024 memory cells.
4. Read the "highs" written during step 3.
5. Complement the test pattern and repeat steps 2, 3 and 4.

Power-up

If $\overline{RAS} = V_{SS}$ during power-up, the KM44C1000B could possibly begin an active cycle. This condition results in higher than necessary current demands from the power supply during power-up. It is recommended that $\overline{RAS}$ and $\overline{CAS}$ track with V_{CC} during power-up or be held at a valid V_{IH} in order to minimize the power-up current.

DEVICE OPERATION (Continued)

An initial pause of 200 μ sec is required after power-up followed by 8 initialization cycles before proper device operation is assured. Eight initialization cycles are also required after any 16 msec period in which there are no RAS cycles. An initialization cycle is any cycle in which RAS is cycled.

Termination

The lines from the TTL driver circuits to the KM44C1000B inputs act like unterminated transmission lines resulting in significant overshoot and undershoot at the inputs. To minimize overshoot it is advisable to terminate the input lines and to keep them as short as possible. Although either series or parallel termination may be used, series termination is generally recommended since it is simple and draws no additional power. It consists of a resistor in series with the input line placed close to the KM44C1000B input pin. The optimum value depends on the board layout. It must be determined experimentally and is usually in the range of 20 to 40 ohms.

Board Layout

It is important to lay out the power and ground lines on memory boards in such a way that switching transient effects are minimized. The recommended methods are gridded power and ground lines or separate power and ground planes. The power and ground lines act like transmission lines to the high frequency transients generated by DRAMS. The impedance is minimized if

all the power supply traces to all the DRAMS run both horizontally and vertically and are connected at each intersection or better yet if power and ground planes are used.

Decoupling

The importance of proper decoupling can not be over emphasized. Excessive transient noise or voltage droop on the V_{CC} line can cause loss of data integrity (soft errors). It is recommended that the total combined voltage changes over time in the V_{CC} to V_{SS} voltage (measured at the device pins) should not exceed 500mV.

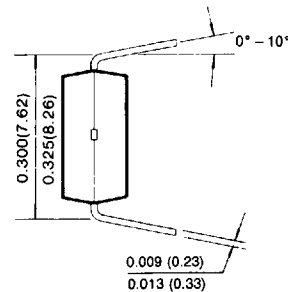
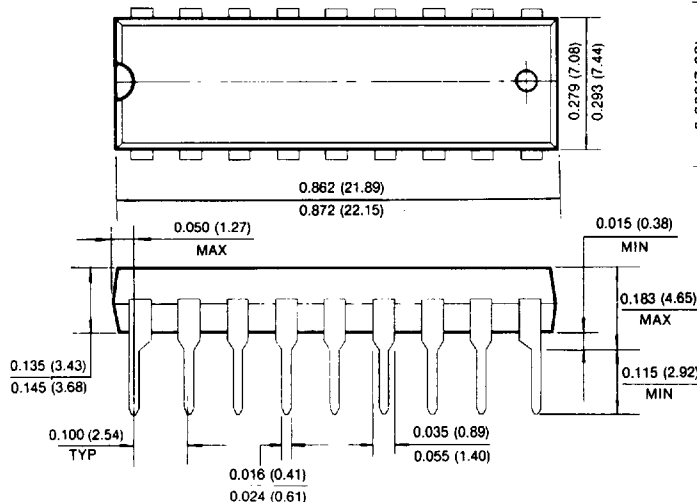
A high frequency 0.1 μ F ceramic decoupling capacitor should be connected between the V_{CC} and ground pins of each KM44C1000B using the shortest possible traces. These capacitors act as a low impedance shunt for the high frequency switching transients generated by the KM44C1000B and they supply much of the current used by the KM44C1000B during cycling.

In addition, a large tantalum capacitor with a value of 47 μ F to 100 μ F should be used for bulk decoupling to recharge the 0.1 μ F capacitors between cycles, thereby reducing power line droop. The bulk decoupling capacitor should be placed near the point where the power traces meet the power grid or power plane. Even better results may be achieved by distributing more than one tantalum capacitor around the memory array.

PACKAGE DIMENSIONS

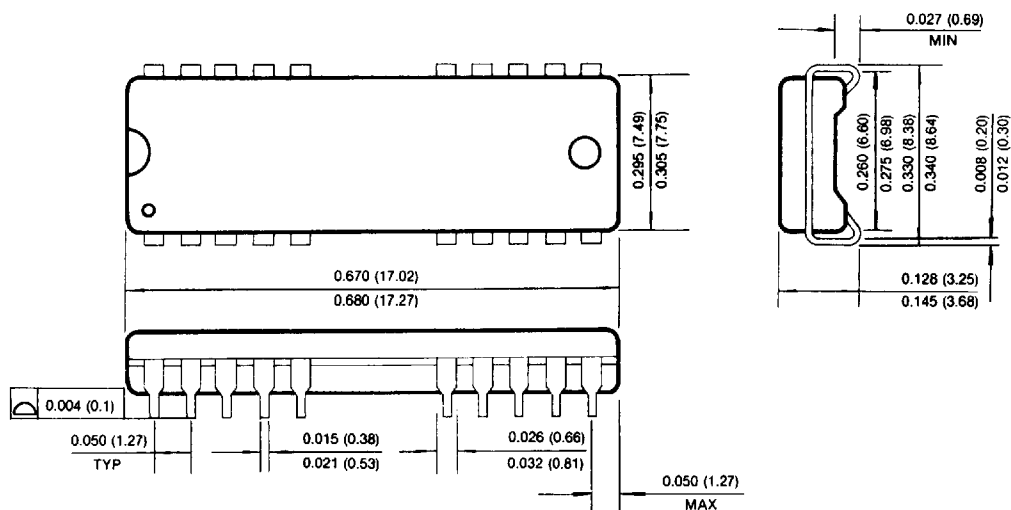
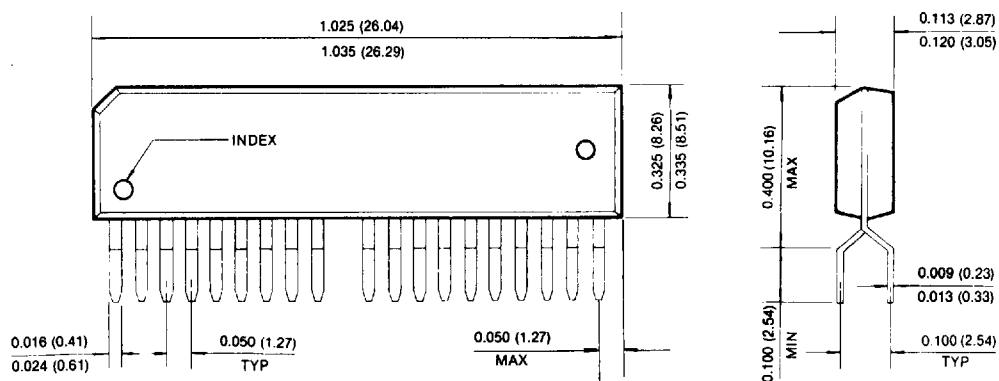
Units: Inches (Millimeters)

18-LEAD PLASTIC DUAL IN-LINE PACKAGE



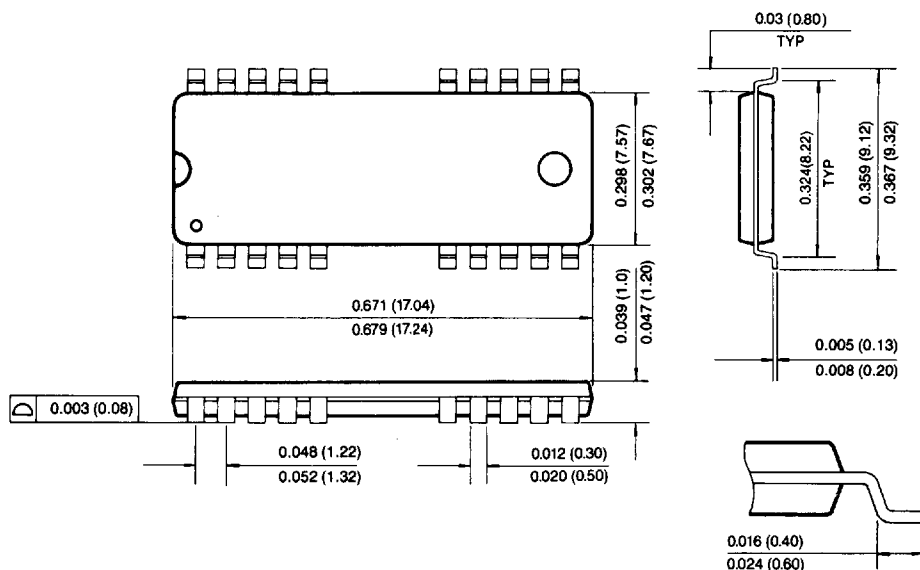
KM44C1000B**CMOS DRAM****PACKAGE DIMENSIONS** (Continued)**20-LEAD PLASTIC SMALL OUT-LINE J-LEAD**

Units: Inches (millimeters)

**20-LEAD PLASTIC ZIGZAG-IN-LINE PACKAGE**

KM44C1000B**CMOS DRAM****PACKAGE DIMENSIONS** (Continued)**20-LEAD PLASTIC THIN SMALL OUT-LINE PACKAGE TYPE (II) (Forward and Reverse Type)**

Units: Inches (millimeters)

**20-LEAD PLASTIC THIN SMALL OUT-LINE PACKAGE TYPE (I) (Forward and Reverse Type)**